Piyusha Akash

Nationality: Sri Lankan | **Phone number:** (+94) 702444916 (Mobile) | **Email address:** contact.piyushaakash@gmail.com | **Website:** https://0x3x0.github.io | **LinkedIn:** Piyusha Akash |

YouTube: InfoSecLK | **Address:** Kelaniya, Sri Lanka, 11300, Sri Lanka (Home)

ABOUT ME

Offensive security enthusiast specializing in reverse engineering, malware engineering, and exploit development. Focused on low-level system behavior, binary analysis, and understanding operating system internals at a deep technical level.

Experienced in C/C++ and x86-64 assembly, with hands-on work in developing custom tools, analyzing compiled binaries, and experimenting with exploitation techniques across Windows and Linux environments. Regularly engages in CTF challenges and practical labs to refine skills in vulnerability discovery and adversarial thinking.

Strong interest in stealthy execution, syscall-level operations, and evasion techniques, with ongoing work in building offensive tooling and researching modern attack methodologies.
Seeking opportunities in security research or offensive engineering roles where deep technical problem-solving and system-level expertise are valued.

SKILLS

Programming Languages

C/C++ | x86-64 Assembly | Bash Scripting

Reverse Engineering & Exploitation

Reverse Engineering | Offensive Tool Engineering | Windows Internals | Malware Analysis

Offensive & Defensive Security

Red Teaming | Penetration Testing | OSINT | Social Engineering | Vulnerability Assessment | OWASP TOP 10 | Wazuh | Splunk | SIEM

Networking

Network Security | Network Troubleshooting

Systems & Platforms

Linux (Debian) | Windows | Kali Linux | Virtualization

PUBLICATIONS

2026

Control Flow Redirection via SetupAPI Callback

Published an offensive security research report analyzing alternative code execution vectors within the Windows operating system internals. Documented a methodology to subvert the native **SetupAPI** (**setupapi.dll**) library **SetupIterateCabinetW** by substituting standard file processing callback routines with pointers to executable memory regions holding test payloads. Explained the technical implementation of bypassing endpoint detection telemetry that heavily monitors conventional execution APIs. Provided a fully functional proof of concept written in **C** along with defensive mitigation strategies highlighting memory backing anomalies for security analysts.

Independent Security Research Report 2026

Authors: Piyusha Akash | **Journal Name**: Self Published Technical Security Research | **Publisher**: Piyusha Akash

Link https://github.com/0x3xp/Native-Adversary-Framework/tree/main/Malware-Dev/Injection/SetupAPI-Callback-Redirection

PROJECTS

2026 – CURRENT

Artemis

Developed **Artemis**, a low level offensive security framework engineered in **C** and **x86 x64 Assembly** for **Windows NT internals** research and endpoint detection evasion. Designed and implemented a modular architecture consisting of three core automation engines driven by a single shared header interface. Built a system service number resolution engine covering over fourteen thousand operating system build entries across Windows 10 and Windows 11. Programmed a specialized assembly stub generator capable of compiling direct inline system calls and indirect system calls to bypass user mode telemetry and mask call stack return addresses. Integrated a native cryptographic library featuring XOR, Rolling XOR, and RC4 ciphers to encrypt generated stubs and protect payloads from memory scanners and static analysis.

Link https://github.com/0x3xp/Artemis

2026 – CURRENT

Native Adversary Framework

Developed **Native Adversary Framework**, a comprehensive research repository focused on low level Windows systems programming and offensive security engineering. Programmed sophisticated payload delivery and process injection vectors using **C and C++** alongside the **Native API** to execute process hollowing and thread context hijacking. Engineered stealth mechanisms using **x64 Assembly** to perform manual **Process Environment Block walking** for dynamic module resolution and implemented custom API hashing to bypass static detection signatures. Built binary analysis tools including custom portable executable parsers for tracking import and export address tables.

Link https://github.com/0x3xp/Native-Adversary-Framework

2026 – 2026

SENTINEX

Developed **SENTINEX**, a lightweight and modular Windows reconnaissance framework engineered for red teamers and security researchers. Programmed the system in **C** to prioritize stealth, minimize system output footprint, and gather critical environment intelligence. Engineered the application to execute low level system discovery operations efficiently while maintaining operational security during authorized penetration testing assessments.

Link https://github.com/0x3xp/SENTINEX

EDUCATION AND TRAINING

15/04/2026 – CURRENT

PRACTICAL LOW LEVEL EXPLOITATION AND REVERSE ENGINEERING PWN college

Engaged in advanced hands on training focused on low level software security binary exploitation and reverse engineering Build custom shellcode payloads and execute sophisticated control flow hijacking techniques including return oriented programming and stack manipulation inside sandboxed environments Develop a deep structural understanding of memory corruption vulnerabilities privilege escalation vectors and operating system core architecture Completed intensive laboratory modules requiring manual reverse engineering of compiled binaries debugging with advanced utilities and bypassing software mitigations

Website www.pwn.college | **Field of study** Information and Communication Technologies

27/12/2024 – CURRENT

PRACTICAL CYBERSECURITY DEVELOPMENT AND COMPETITIVE HACKING TryHackMe

Engaged in continuous hands on technical skill development covering advanced penetration testing reverse engineering and threat hunting Build and test exploit mechanics within highly competitive lab environments Achieved consecutive first place rankings in weekly global performance leagues by rapidly solving complex challenge rooms Demonstrated practical proficiency in active directory exploitation memory forensics and vulnerability chaining across hardened Linux and Windows nodes Automated enumeration tasks using custom scripting and terminal utilities to maximize efficiency during time sensitive security challenges

Website www.tryhackme.com | **Field of study** Information and Communication Technologies

CERTIFICATIONS

Hackviser, 24/02/2026

Certified Associate Penetration Tester

Practical hands on certification validating core competencies in ethical hacking network exploitation web application security and privilege escalation Proves operational readiness via a rigorous practical examination requiring vulnerability chaining network reconnaissance and professional reporting

Mode of learning: Online

Link https://hackviser.com/verify?id=HV-CAPT-HKXRNSLE

Red Team Leaders, 2026

Certified Windows Offensive API

Advanced offensive engineering certification focusing on native Windows internals low level development and Win32 API manipulation Validates the ability to bypass user land hooks using direct system calls and build custom toolsets in C and C++ Cleared a hands on 24 hour practical examination

Mode of learning: Research-Lab based

Link https://linkedin.com/in/piyushaakash

The Linux Foundation, 2026

Introduction to Linux LFS101

Comprehensive training covering Linux system administration command line architecture and process management Validates operational proficiency in file system navigation user permissions data manipulation tools and automated Bash shell scripting

Mode of learning: Online

Link https://linkedin.com/in/piyushaakash

CISCO, 2024

Introduction to Cybersecurity

Foundational course covering core security concepts threat vectors and network infrastructure protection Validates understanding of the confidentiality integrity and availability triad alongside basic device hardening principles

Mode of learning: Online

Link https://linkedin.com/in/piyushaakash

LANGUAGE SKILLS

Mother tongue(s): **SINHALA**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	A2	A1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user